Лабораторная работа

Номер 4

Кобзев Д. К.

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Российский университет дружбы народов, Москва, Россия

Информация

Докладчик

- Кобзев Дмитрий Константинович
- Студент
- Российский университет дружбы народов
- НПИбд-01-23

Цель работы

Целью данной работы является проведение подготовительной работы по первоначальной настройке коммутаторов сети.

Первоначальное конфигурирование сети

В логической рабочей области Packet Tracer размещаем коммутаторы и оконечные устройства согласно схеме сети L1 и соединяем их через соответствующие интерфейсы (Рис. 1.1).

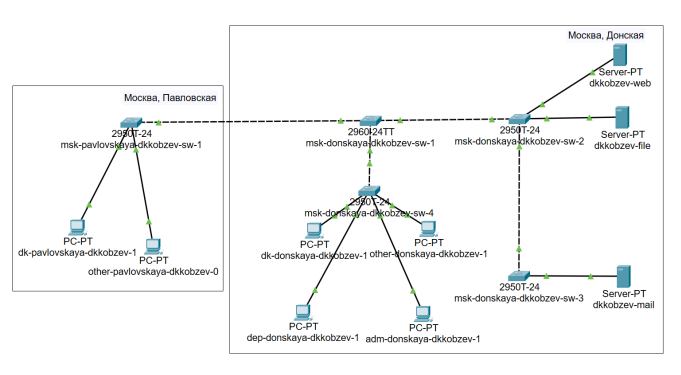

Рис. 1: Размещение коммутаторов и оконечных устройств согласно схеме сети L1

Первоначальное конфигурирование сети

Используя типовую конфигурацию коммутатора, настраиваем все коммутаторы, изменяя название устройства и его IP-адрес согласно плану IP (Рис. 1.2), (Рис. 1.3), (Рис. 1.4), (Рис. 1.5), (Рис. 1.6).

```
Switch>enable
Switch# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname msk-donskaya-dkkobzev-sw-1
msk-donskaya-dkkobzev-sw-1(config)#interface vlan2
msk-donskaya-dkkobzev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-dkkobzev-sw-1(config-if)#no shutdown
msk-donskaya-dkkobzev-sw-1(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-dkkobzev-sw-1(config-if)#exit
msk-donskaya-dkkobzev-sw-1(config)#ip default  gateway 10.128.1.1
                                               ^
% Invalid input detected at '^' marker.

msk-donskaya-dkkobzev-sw-1(config)#ip default-gateway 10.128.1.1
msk-donskaya-dkkobzev-sw-1(config)#line vty 0 4
msk-donskaya-dkkobzev-sw-1(config-line)#password cisco
msk-donskaya-dkkobzev-sw-1(config-line)#login
msk-donskaya-dkkobzev-sw-1(config-line)#exit
msk-donskaya-dkkobzev-sw-1(config)# line console 0
msk-donskaya-dkkobzev-sw-1(config-line)#password cisco
msk-donskaya-dkkobzev-sw-1(config-line)#login
msk-donskaya-dkkobzev-sw-1(config-line)#exit
msk-donskaya-dkkobzev-sw-1(config)#enable secret cisco
msk-donskaya-dkkobzev-sw-1(config)# service password-encryption
msk-donskaya-dkkobzev-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-dkkobzev-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-dkkobzev-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dkkobzev-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dkkobzev-sw-1(config)#line vty 0 4
*Mar 1 0:29:23.676: %SSH Key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:29:23.676: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dkkobzev-sw-1(config-line)# transport input ssh
msk-donskaya-dkkobzev-sw-1(config-line)#exit
msk-donskaya-dkkobzev-sw-1(config)#exit
msk-donskaya-dkkobzev-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dkkobzev-sw-1# write memory
Building configuration...
[OK]
```

Рис. 2: Первоначальная настройка коммутатора msk-donskaya-dkkobzev-sw-1

Первоначальное конфигурирование сети

```
Switch>enable
Switch# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname msk-donskaya-dkkobzev-sw-2
msk-donskaya-dkkobzev-sw-2(config)# interface vlan2
msk-donskaya-dkkobzev-sw-2(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-dkkobzev-sw-2(config-if)#no shutdown
msk-donskaya-dkkobzev-sw-2(config-if)#ip address 10.128.1.3 255.255.255.0
msk-donskaya-dkkobzev-sw-2(config-if)#exit
msk-donskaya-dkkobzev-sw-2(config)#ip default  gateway 10.128.1.1
                                               ^
% Invalid input detected at '^' marker.

msk-donskaya-dkkobzev-sw-2(config)#ip default-gateway 10.128.1.1
msk-donskaya-dkkobzev-sw-2(config)#line vty 0 4
msk-donskaya-dkkobzev-sw-2(config-line)#password cisco
msk-donskaya-dkkobzev-sw-2(config-line)#login
msk-donskaya-dkkobzev-sw-2(config-line)#exit
msk-donskaya-dkkobzev-sw-2(config)# line console 0
msk-donskaya-dkkobzev-sw-2(config-line)#password cisco
msk-donskaya-dkkobzev-sw-2(config-line)#login
msk-donskaya-dkkobzev-sw-2(config-line)#exit
msk-donskaya-dkkobzev-sw-2(config)#enable secret cisco
msk-donskaya-dkkobzev-sw-2(config)## service password-encryption

% Invalid input detected at '^' marker.

msk-donskaya-dkkobzev-sw-2(config)# service password-encryption
msk-donskaya-dkkobzev-sw-2(config)#username admin privilege 1 secret cisco
msk-donskaya-dkkobzev-sw-2(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-dkkobzev-sw-2(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dkkobzev-sw-2.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dkkobzev-sw-2(config)#line vty 0 4
*Mar 1 0:35:36.124: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:35:36.124: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dkkobzev-sw-2(config-line)# transport input ssh
msk-donskaya-dkkobzev-sw-2(config-line)#exit
msk-donskaya-dkkobzev-sw-2(config)#exit
msk-donskaya-dkkobzev-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dkkobzev-sw-2#write memory
Building configuration...
[OK]
```

Рис. 3: Первоначальная настройка коммутатора msk-donskaya-dkkobzev-sw-2

Первоначальное конфигурирование сети

```
Switch>enable
Switch# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname msk-donskaya-dkkobzev-sw-3
msk-donskaya-dkkobzev-sw-3(config)#interface vlan2
msk-donskaya-dkkobzev-sw-3(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-dkkobzev-sw-3(config-if)#no shutdown
msk-donskaya-dkkobzev-sw-3(config-if)#ip address 10.128.1.4 255.255.255.0
msk-donskaya-dkkobzev-sw-3(config-if)#exit
msk-donskaya-dkkobzev-sw-3(config)#ip default-gateway 10.128.1.1
msk-donskaya-dkkobzev-sw-3(config)# line vty 0 4
msk-donskaya-dkkobzev-sw-3(config-line)#password cisco
msk-donskaya-dkkobzev-sw-3(config-line)#login
msk-donskaya-dkkobzev-sw-3(config-line)#exit
msk-donskaya-dkkobzev-sw-3(config)#line console 0
msk-donskaya-dkkobzev-sw-3(config-line)#password cisco
msk-donskaya-dkkobzev-sw-3(config-line)#login
msk-donskaya-dkkobzev-sw-3(config-line)#exit
msk-donskaya-dkkobzev-sw-3(config)#enable secret cisco
msk-donskaya-dkkobzev-sw-3(config)# service password-encryption
msk-donskaya-dkkobzev-sw-3(config)#username admin privilege 1 secret cisco
msk-donskaya-dkkobzev-sw-3(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-dkkobzev-sw-3(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dkkobzev-sw-3.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dkkobzev-sw-3(config)#line vty 0 4
*Mar 1 0:38:27.933: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:38:27.933: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dkkobzev-sw-3(config-line)# transport input ssh
msk-donskaya-dkkobzev-sw-3(config-line)#exit
msk-donskaya-dkkobzev-sw-3(config)#exit
msk-donskaya-dkkobzev-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dkkobzev-sw-3#write memory
Building configuration...
[OK]
```

Рис. 4: Первоначальная настройка коммутатора msk-donskaya-dkkobzev-sw-3

Первоначальное конфигурирование сети

```
Switch>enable
Switch# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname msk-donskaya-dkkobzev-sw-4
msk-donskaya-dkkobzev-sw-4(config)#interface vlan2
msk-donskaya-dkkobzev-sw-4(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-dkkobzev-sw-4(config-if)#no shutdown
msk-donskaya-dkkobzev-sw-4(config-if)#ip address 10.128.1.5 255.255.255.0
msk-donskaya-dkkobzev-sw-4(config-if)#exit
msk-donskaya-dkkobzev-sw-4(config)#ip default  gateway 10.128.1.1
                                               ^
% Invalid input detected at '^' marker.

msk-donskaya-dkkobzev-sw-4(config)#ip default-gateway 10.128.1.1
msk-donskaya-dkkobzev-sw-4(config)#line vty 0 4
msk-donskaya-dkkobzev-sw-4(config-line)# password cisco
msk-donskaya-dkkobzev-sw-4(config-line)#login
msk-donskaya-dkkobzev-sw-4(config-line)#exit
msk-donskaya-dkkobzev-sw-4(config)#line console 0
msk-donskaya-dkkobzev-sw-4(config-line)# password cisco
msk-donskaya-dkkobzev-sw-4(config-line)#login
msk-donskaya-dkkobzev-sw-4(config-line)#exit
msk-donskaya-dkkobzev-sw-4(config)#enable secret cisco
msk-donskaya-dkkobzev-sw-4(config)# service password  encryption
                                                      ^
% Invalid input detected at '^' marker.

msk-donskaya-dkkobzev-sw-4(config)# service password-encryption
msk-donskaya-dkkobzev-sw-4(config)#username admin privilege 1 secret cisco
msk-donskaya-dkkobzev-sw-4(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-dkkobzev-sw-4(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dkkobzev-sw-4.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dkkobzev-sw-4(config)#line vty 0 4
*Mar 1 0:32:10.407: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:32:10.407: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dkkobzev-sw-4(config-line)#transport input ssh
msk-donskaya-dkkobzev-sw-4(config-line)#exit
msk-donskaya-dkkobzev-sw-4(config)#exit
msk-donskaya-dkkobzev-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dkkobzev-sw-4# write memory
Building configuration...
[OK]
```

Рис. 5: Первоначальная настройка коммутатора msk-donskaya-dkkobzev-sw-4

Первоначальное конфигурирование сети

```
Switch(config)#hostname msk-pavlovskaya-dkkobzev-sw-1
msk-pavlovskaya-dkkobzev-sw-1(config)#interface vlan2
msk-pavlovskaya-dkkobzev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-pavlovskaya-dkkobzev-sw-1(config-if)#no shutdown
msk-pavlovskaya-dkkobzev-sw-1(config-if)#ip address 10.128.1.6 255.255.255.0
msk-pavlovskaya-dkkobzev-sw-1(config-if)#exit
msk-pavlovskaya-dkkobzev-sw-1(config)#ip default  gateway 10.128.1.1
                                                  ^
% Invalid input detected at '^' marker.

msk-pavlovskaya-dkkobzev-sw-1(config)#ip default-gateway 10.128.1.1
msk-pavlovskaya-dkkobzev-sw-1(config)# line vty 0 4
msk-pavlovskaya-dkkobzev-sw-1(config-line)#password cisco
msk-pavlovskaya-dkkobzev-sw-1(config-line)#login
msk-pavlovskaya-dkkobzev-sw-1(config-line)#exit
msk-pavlovskaya-dkkobzev-sw-1(config)# line console 0
msk-pavlovskaya-dkkobzev-sw-1(config-line)#password cisco
msk-pavlovskaya-dkkobzev-sw-1(config-line)#login
msk-pavlovskaya-dkkobzev-sw-1(config-line)#exit
msk-pavlovskaya-dkkobzev-sw-1(config)#enable secret cisco
msk-pavlovskaya-dkkobzev-sw-1(config)# service password-encryption
msk-pavlovskaya-dkkobzev-sw-1(config)#username admin privilege 1 secret cisco
msk-pavlovskaya-dkkobzev-sw-1(config)#ip domain name donskaya .rudn.edu
                                                     ^
% Invalid input detected at '^' marker.

msk-pavlovskaya-dkkobzev-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-pavlovskaya-dkkobzev-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-pavlovskaya-dkkobzev-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-dkkobzev-sw-1(config)#line vty 0 4
*Mar 1 0:23:30.125: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:23:30.125: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-pavlovskaya-dkkobzev-sw-1(config-line)# transport input ssh
msk-pavlovskaya-dkkobzev-sw-1(config-line)#exit
msk-pavlovskaya-dkkobzev-sw-1(config)#exit
msk-pavlovskaya-dkkobzev-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-pavlovskaya-dkkobzev-sw-1#write memory
Building configuration...
[OK]
```

Рис. 6: Первоначальная настройка коммутатора msk-donskaya-dkkobzev-sw-1

Выводы

В результате выполнения лабораторной работы мною была проведена подготовительная работа по первоначальной настройке коммутаторов сети.